IoT-Core Assistant

SW-Version: 1.1

Short description

The IoT-Core Assistant provides following functions:

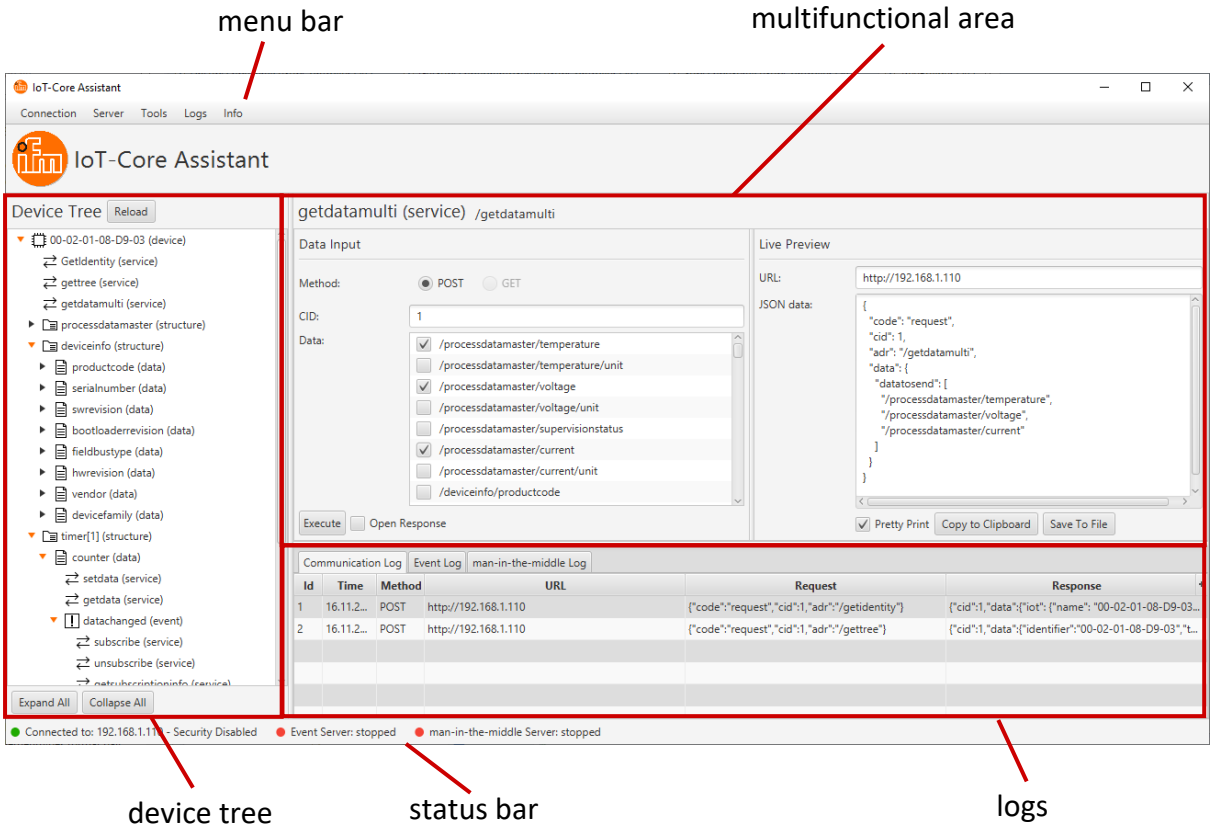
- Connect to IoT-Core devices, with and without enabled security
- Display the complete and interactive device tree of the connected IoT-Core device
- Build and send IoT-Core service requests via POST or GET method
    - Native support for many services
    - Generic support for unknown services
    - Live preview of JOSN service requests
- Read offline device tree from a local JSON file (services can't be executed)
- Event server to receive events from IoT-Core devices
- Man-in-the-middle server to record the HTTP communication between a client and an IoT-Core device
- Send generic HTTP-request to arbitrary device
- Separate ring buffer to record and log:
    - Communication to the connected device
    - Received IoT-Core events by the event server
    - Forwarded communication by the man-in-the-middle server
- Export each log to a CSV file

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1. Overview of the user interface

The central UI component of the IoT-Core Assistant is the Main Window. From here all functionalities of the application can be accessed and controlled. The window itself is divided into multiple segments. The following section will introduce all of these. More detailed information of the functions can be found in the following chapters.



menu bar

The menu bar is in the top of the window and gives access to all the application functionalities and features. Following functions can be access from here:

- Connection: Connect to new device or loading an offline device from a file
- Server: Controlling the event and man-in-the-middle server
- Tools: Sending HTTP requests independent from the connected device
- Logs: Save or clear the communication, event, and man-in-the-middle log
- Info: Information of the current application

multifunctional area

The multifunctional area is the center part of the whole window. The displayed content depends on the user selection and the state of the application.

device tree
The device tree of the connected IoT-Core device is shown on the left side of the screen. This is an interactive representation of the device, with all its data and functionalities listed. Each of the five IoT-Core elements has its own icon:

icon	element type	description
	device	Represent the IoT-Core device and is always the root element of the entire device tree
	structure	Used to structure the device tree. Equivalent to a folder in an operation system
	data	Represents a data point in the IoT-Core device
	service	Represents an executable service in the IoT-Core device
	event	Represents an event node, that can be subscribe to and will send a http request on change to all subscribers

For easier use the entire device tree can be expanded or collapsed with the buttons *Expand All* and *Collapse All*. The button *Reload* can be used to reload the device tree of the connected device without entering the address again.

Status bar
The status bar is located at the bottom of the window and shows the status of the connected device and both servers.

Logs
All recorded communication, event server and man-in-the-middle server log entries will be displayed in a separate list under the main content.

2. Connect to new device

By connecting to a new IoT-Core device the IoT-Core Assistant will get all necessary information from the device to access all its functionalities. This includes security information and the device tree containing every accessible datapoint and functionality. When the connection was successful the corresponding device tree will be displayed on the left side of the main window.

Connect to new device

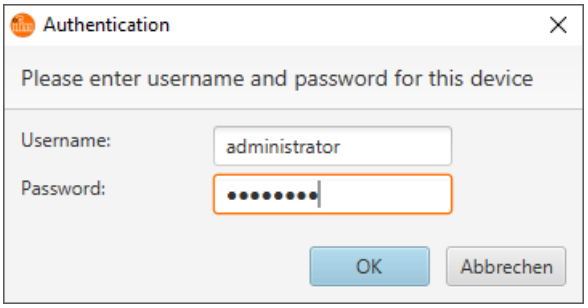
To connect to a new IoT-Core device, please select from the menu bar:

Connection → New Connection

The *New Connection* dialog will open. Please enter the IP address and the Port of the desired IoT-Core device. When no Port is specified, Port 80 will be used by default. To connect to the device click *Connect*.



The IoT-Core Assistant will try to reach the specified device and get its identity information. If the device cannot be reached, an error message is displayed. When the security mode of the device is activated, further authentication information is needed. In this case the username and password must be entered in the authentication dialog. This step will be skipped when no authentication is needed.



When all required data have been entered and the device has been successfully found on the network, the device description will be loaded and displayed in the device tree. Additionally, the status will be set to connected.

3. Load device tree from file

A device tree can be loaded from a local previously exported JSON file. By loading a device tree from a local file, most functionalities provided by the IoT-Core Assistant can be used without a present IoT-Core device. However, build IoT-Core requests can't be executed. This can be useful, when the desired IoT-Core device isn't available now.

A valid device tree file contains the raw response data from a *gettree* service request. Exporting and loading this file is described in the following.

Export device tree file

To export a device tree file, first connect to an IoT-Core device. Select the automatically executed *gettree* service from the Communication Log. Save the JSON response of the *gettree* service in a file, by clicking *Save to File.*

Load tree from file

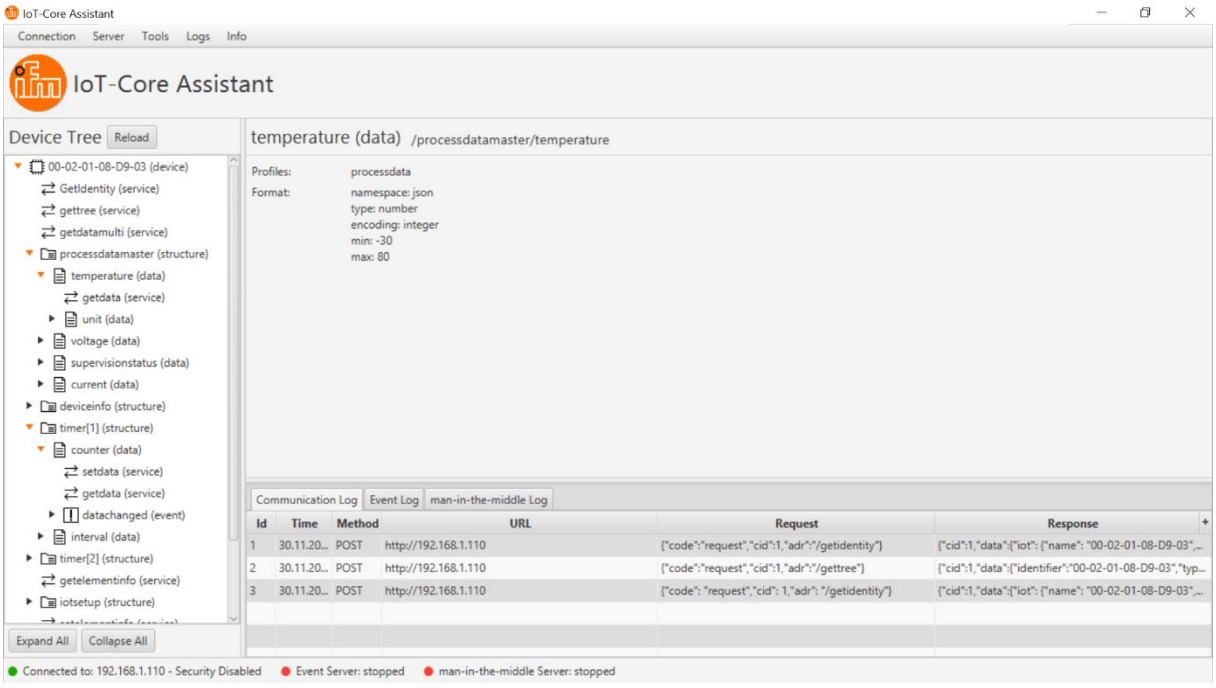
To load a device tree from a local file, please select from the menu bar:

Connection → Load Tree from File

Please select the previously exported JSON file from the file dialog. After confirmation, the device tree is read from the selected file. In the status bar a device tree loaded from a file is displayed as an offline device.

4. Show detailed information of IoT-Core Elements

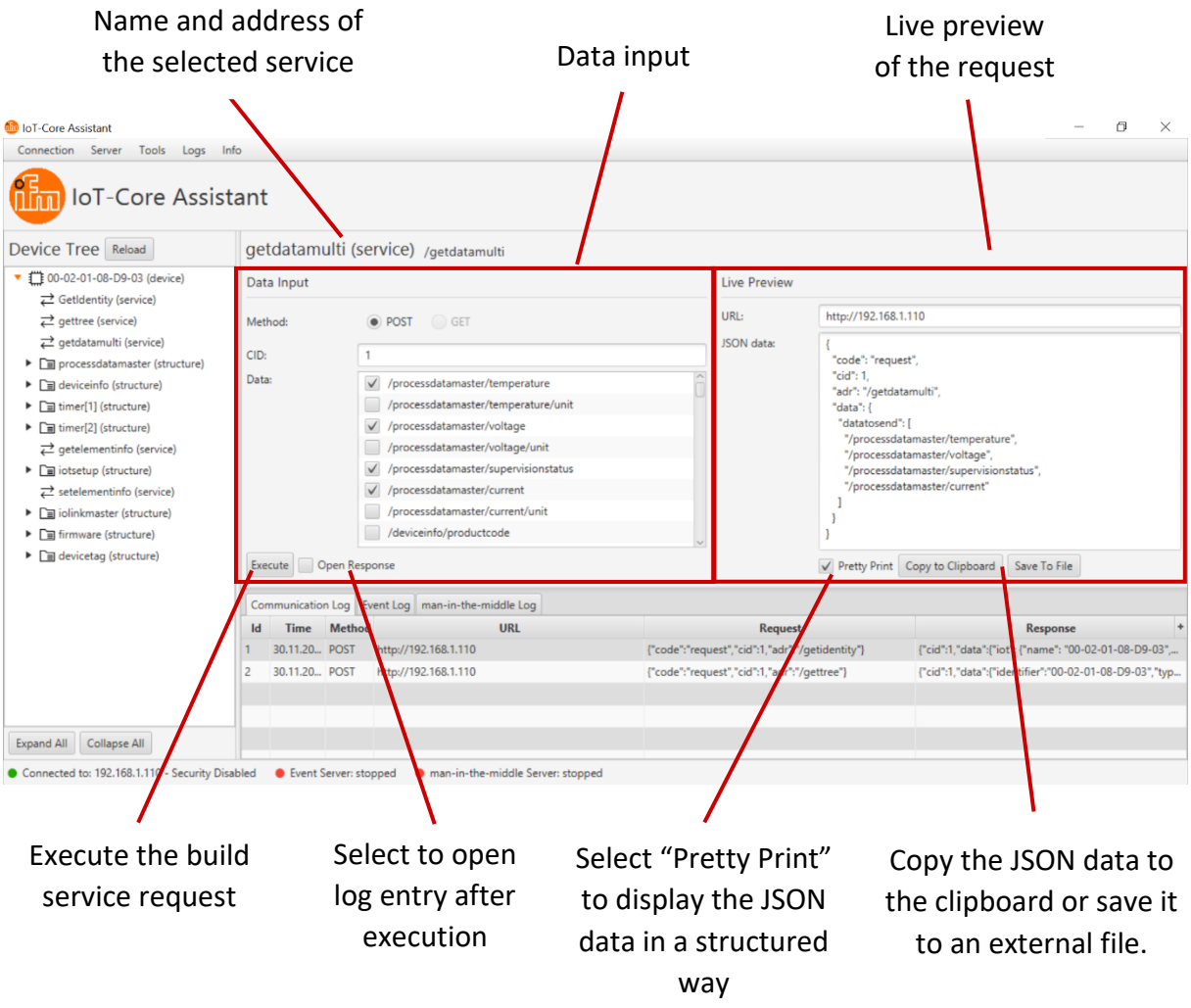
By selecting an element from the device tree, its detailed information will be displayed in the center of the main window. Supported IoT-Core Elements are device, structure, data and event.



5. Build and execute IoT-Core service requests

By selecting a service from the device tree, an interactive interface for generating the service request is shown in the centered multifunction area. All parameters required for this service can be entered on the left. The corresponding JSON data for this service is automatically updated on the right. The data structures and input fields may vary depending on the selected service. For unsupported services the basic structure will be identical. However, the required data must be provided by the user in a valid JSON format. Invalid or empty JSON data will be ignored. A list of all native supported services by the IoT-Core Assistant can be found in 5.1.

IoT-Core services can be send using two methods POST and GET. The POST method is the default method and is supports by all services. A POST request will always require a JSON Object with the information. The GET method is only available when the selected service doesn't require any data send to the IoT-Core device and no authentication is needed. Then the path of the selected service is sent in the URL of the request.



5.1. Natively supported services

The following services are natively supported by the IoT-Core Assistant:

- factoryreset
- getblock
- getdatamulti
- getdata
- getelementinfo
- getidentity
- getsubscriptioninfo
- gettree
- iolreadacyclic
- iolwriteacyclic
- projectall
- reboot
- reseterrorcounter
- setblock (With all supported datatypes by setdata)
- setdata
    - Supported datatypes:
        - array
        - boolean
        - enum (Encoding: integer)
        - enum (Encoding: string)
        - number (Encoding: double)
        - number (Encoding: float)
        - number (Encoding: integer)
        - object
        - string
- signal
- subscribe
- unsubscribe
- validationuseconnecteddevice

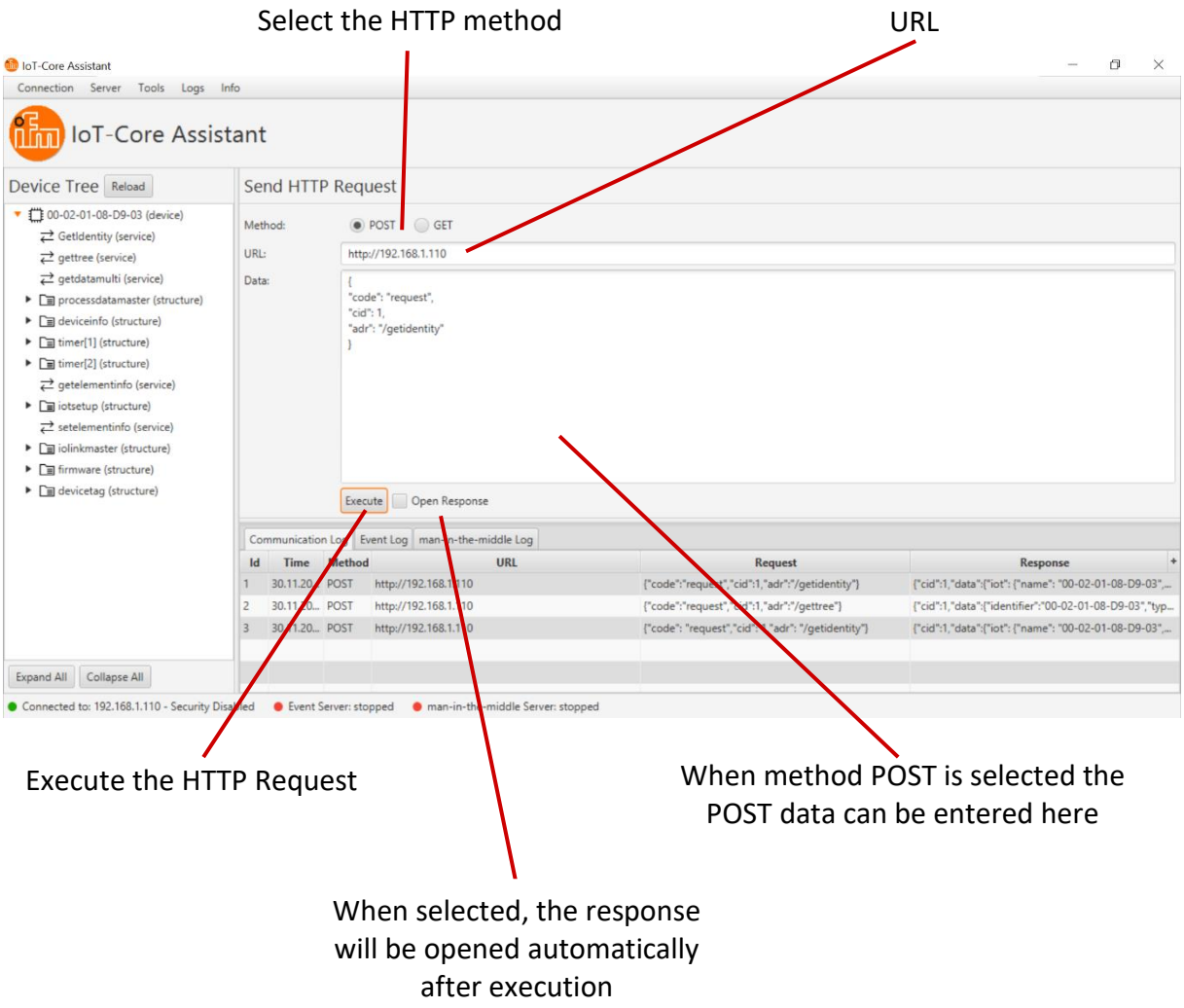
6. Send independent HTTP requests

Send HTTP Request is a built-in tool that can be used to send independent http requests to any device. This represents a simple HTTP client and allows the user to send any HTTP/HTTPS request. The communication will be saved in the communication log.

To open the tool, please select from the menu bar:

Tools → Send HTTP Request

This opens the following user interface:



7. Event server

The event server can be used to receive events from one or multiple IoT-Core devices. The server will listen to one Port and will log all received events. This port can be changed when starting the event server. All received events are recorded in the event log. To see detailed information of the received event, select the event from the log table.

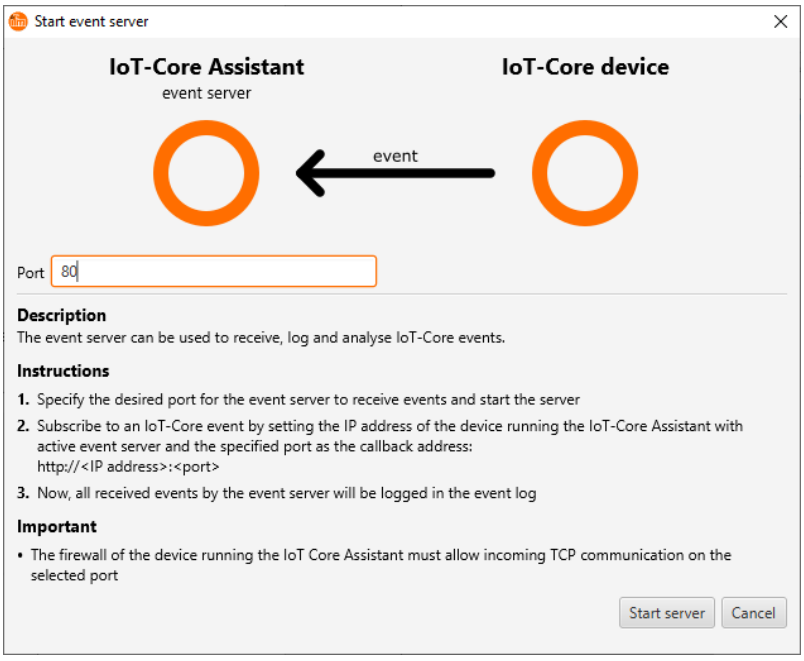
NOTE: Java needs permission to use the specified TCP-Port.

Start the event server

To start the event server, please select from the menu bar:

Server → Start Event server

The following dialog containing another short description and instructions will open. Please enter the port for the event server to receive events on and start the server by clicking start server



When the event server was started successfully, the status of the event server will be symbolized by a green dot in the status bar.

Stop the event server

To stop a running event server, please select in the menu bar:

Server → Stop event server

This option is only possible when the event server is running.

Using the event server

When running, the event server will record every http communication received on the defined port while starting the server. The receiving port can be checked in the status bar. When the received data are a valid IoT-Core event, the event server will extract all important information and store these with the raw data in the event log. When the data aren't a valid IoT-Core event, only the raw data will be stored, and the entry will be marked as invalid.

When an event in an IoT-Core device fires, every subscribed ip address will be notified with an IoT-Core event. The service *subscribe* will subscribe one device to on event.

8. Man-in-the-middle server

The man-in-the-middle server can be used to record and log communication between arbitrary network devices and a device supporting the IoT-Core communication protocol. This is achieved by executing a man-in-the-middle attack. To do this, all service requests must be sent to the IP address of the local machine. All requests are then forwarded to a freely selectable IP address. The subsequent response is also routed through the man-in-the-middle server. This allows the entire communication to be recorded.

NOTE: Only HTTP protocol is supported.

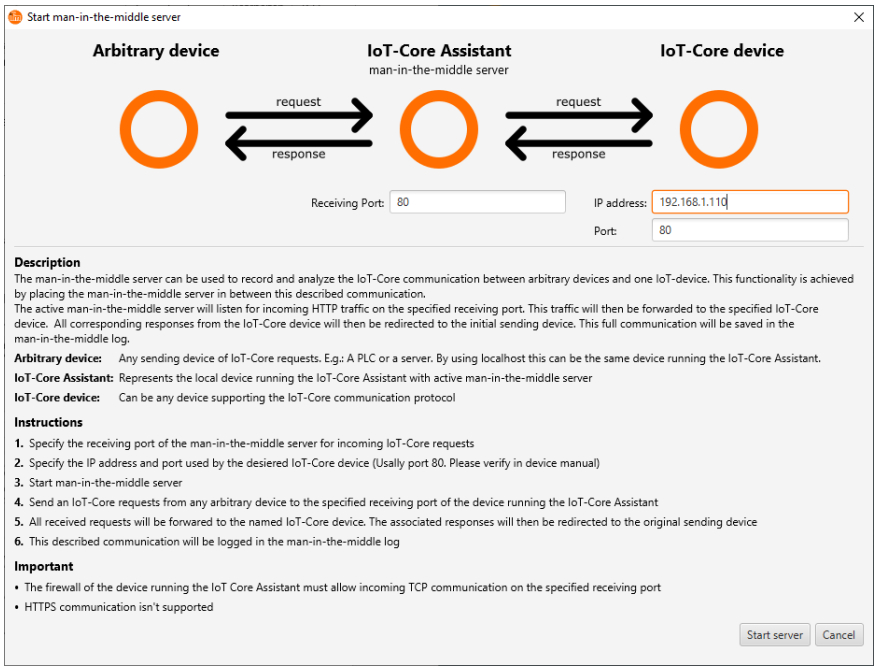
NOTE: Java needs permission to use the receiving TCP-Port.

<u>Start man-in-the-middle server</u>

To start the server, please select from the menu bar:

Server → Start man-in-the-middle server

This opens a dialog in which the IP address of the IoT-Core device must be entered.



NOTE: The man-in-the-middle Serve uses port 80. Therefore port 80 cannot be used while the man-in-the-middle server is running.

In the status bar a running server is symbolized by a green dot. When the server is running, all http communication received at port 80 is forwarded to the specified IP address. This communication will be recorded in the man-in-the-middle protocol.

Stop man-in-the-middle server

To stop the server, please select in the menu bar:

Server → Stop man-in-the-middle server

This option is only possible when the man-in-the-middle server is running.

Using the event server

When running, the man-in-the-middle server will listen on the specified receiving port for http requests. These will then be forwarded to the given IoT-Core device. Answers from the IoT-Core device will then be redirected to the initial sending device. All data from the request and response will be recorded and will be saved in the man-in-the-middle log. For the sending device, the communication isn't any different than the direct connection to the IoT-Core device. The timeout for the http request is set to 10 seconds. On an occurring timeout the log entry will only save the request data and be marked as invalid.

9. Logs

The IoT-Core Assistant has three different types of logs:

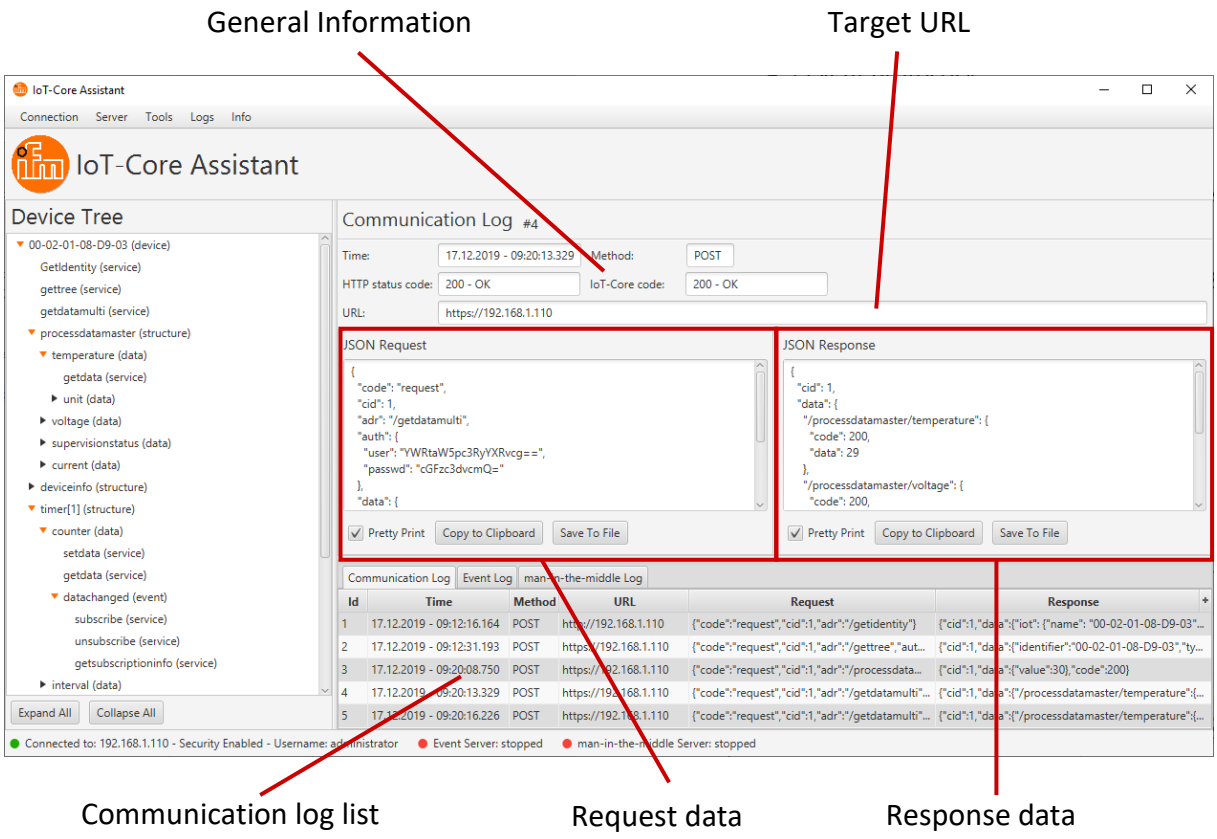
- Communication log
- Event log
- Man-in-the-middle log

These are listed individually in the bottom of the main window. By selecting an entry, the corresponding detail information will be shown in the multifunctional display. An entry which is highlighted in red indicates invalid data or a timeout while sending.

Note: Each protocol can store a maximum of 200 entries. Older entries will be overwritten!

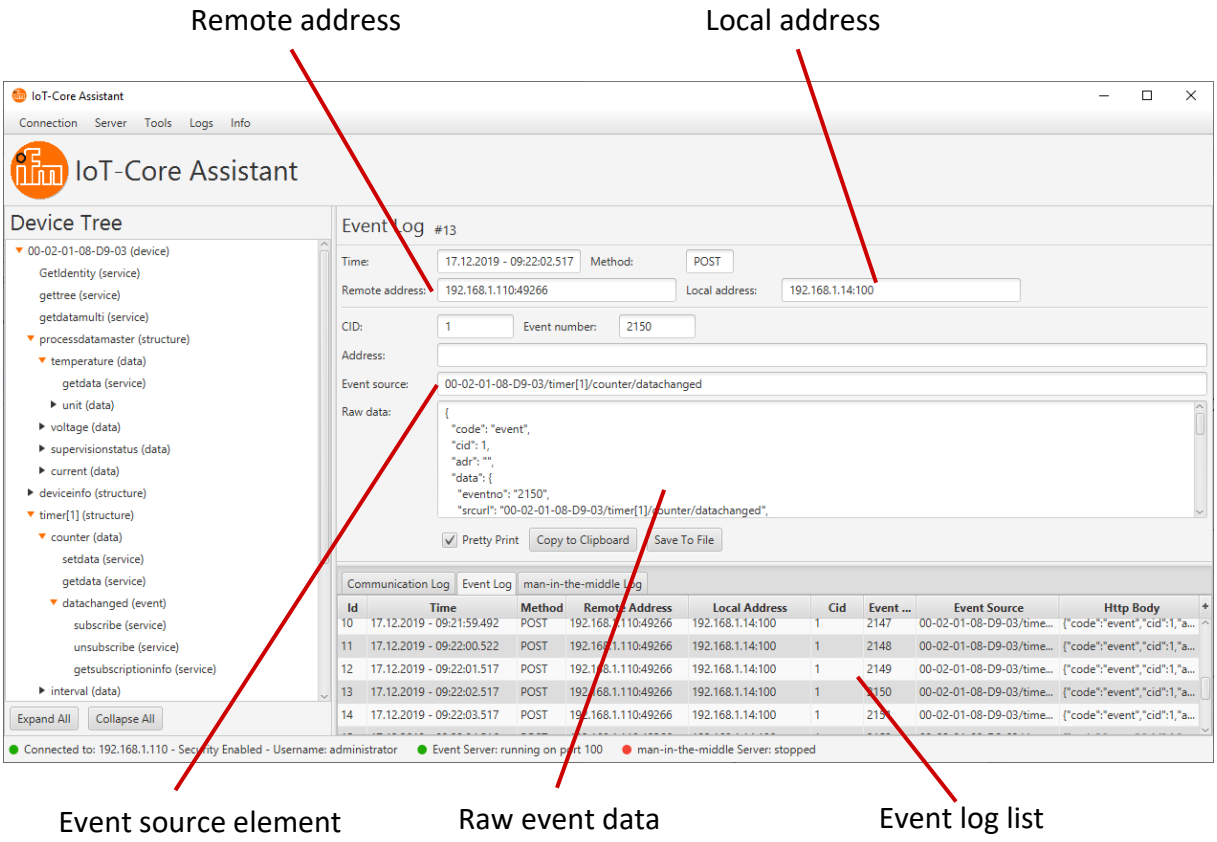
9.1. Communication Log

The communication log saves all requests that are executed from the IoT-Core Assistant. This includes requests to the connected IoT-Core device and independent HTTP requests.



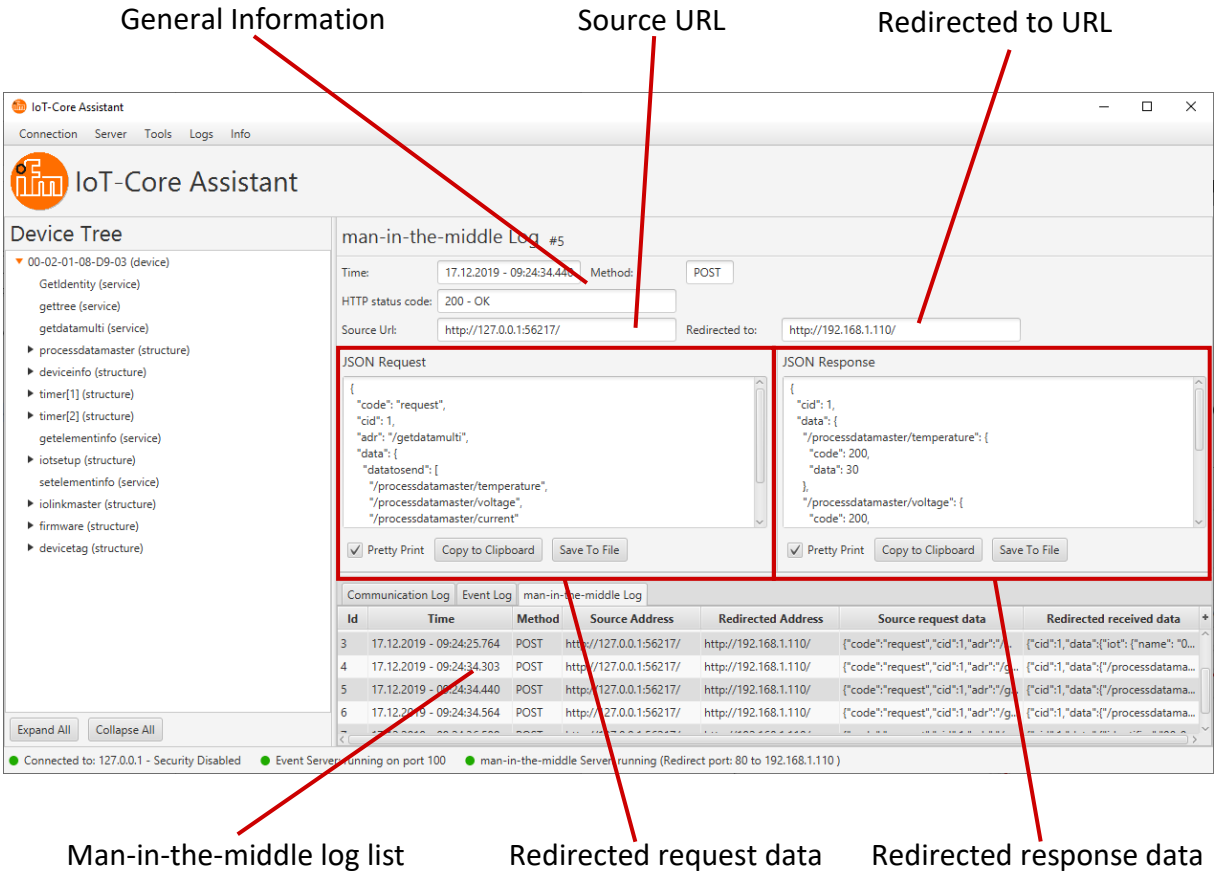
9.2. Event Log

The event log saves all IoT-Core events received by the event server.



9.3. Man-in-the-middle Log

The man-in-the-middle log saves every communication forwarded by the man-in-the-middle server.



9.4. Export Log

Each protocol can be exported individually to a CSV file. To export a protocol, please select in the menu bar:

Logs → [Protocol type] → Save to File

This will open a dialog to select the storage location of the CSV file. After selecting the desired location, the protocol will be exported to the CSV file.

9.5. Clear Log

Each log can be cleared individually. This will delete every entry in the log. To clear a log, please select in the menu bar:

Logs → [Log type] → Clear

10. Possible Issues

Issues	**Possible solution**
Running event server doesn't receive events	Java needs permission to use the chosen TCP port. Please check your firewall settings.
Running man-in-the-middle server doesn't forward messages	Java needs permission to use the TCP port 80. Please check your firewall settings.

11. Version History

Version: 1.1 – 2020/11/23

#Added

- Added native support for services:
    - setblock
    - getblock
- Ability to specify ports for man-in-the-middle server
- More explaining dialogs when starting man-in-the-middle server and event server
- Added icons for the different IoT-Core element types in the device tree view
- Added the option in setdata service to automatically encode new value string with base64, when encoding of the data point is base64
- Ability to reload device tree from connected device
- Updated IoT-Core error codes

#Fixed

- Improved UI handling when resizing the application and using display scaling factors other than 100%
- Fixed Bug in the new Connection Dialog that causes the connect button to stay enabled when port was invalid
- Fixed Bug that causes the application to not close properly
- Fixed Bug with handling incoming events that are in an invalid JSON-Format
- Fixed Bug where new data in the unknown service builder won't validate properly
- Fixed Bug where error messages had no window icon
- Logged data containing text with multiple lines will now be displayed without page breaks in all log table columns
- Disabled GET-Method for subscribe and unsubscribe service
- Spelling errors

Version: 1.0.1 - 2020/08/10

#Added

- Ability to specify port when establishing new connection

#Fixed

- When an error occurred while connection to a new device, the connect button sometimes was stuck on abort